Quadratic residues

L5-NT1 Level 5 Advanced · Number theory

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November 2026

Legendre and Jacobi symbols, Euler's criterion, the laws for -1 and 2, quadratic reciprocity, Thue's lemma, sums of two squares and counting with the symbol.
By the end you can evaluate Legendre and Jacobi symbols with Euler's criterion and quadratic reciprocity; determine which primes divide values of a quadratic, and use this to rule out solutions of equations; use Thue's lemma, the two squares theorem and sums of Legendre symbols in olympiad proofs.
Prerequisites: Orders, Fermat and Euler (L4-NT1), Olympiad Diophantine equations (L4-NT2).
Contests: BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Let n be odd and let p be a prime factor of 2^n+1. Orders (L4-NT1) only give $\operatorname{ord}_p(2) \mid 2n$, which says nothing about p modulo 8. Instead multiply $2^n \equiv -1$ by 2. Since $n+1$ is even,

$$-2 \equiv 2^{n+1} = \left(2^{(n+1)/2}\right)^2 \pmod{p},$$

so -2 is a square modulo p. Theorem 2.2 below shows that this happens only for $p \equiv 1$ or $3 \pmod 8$. For example $2^5+1 = 3 \cdot 11$ and $2^7+1 = 3 \cdot 43$, and $3, 11, 43$ are all 3 modulo 8. This unit decides, for any integer a, the primes modulo which a is a square.

Throughout, p is an odd prime. Orders, Lagrange's theorem on polynomial congruences and the Chinese remainder theorem (L4-NT1, L4-NT2) are assumed.

1.1 Residues and the Legendre symbol

An integer a with $p \nmid a$ is a **quadratic residue** modulo p if $x^2 \equiv a \pmod p$ has a solution, and a **non-residue** otherwise. The **Legendre symbol** $\left(\frac{a}{p}\right)$ is 1 for a residue, -1 for a non-residue and 0 if $p \mid a$. With this convention $x^2 \equiv a$ has exactly $1 + \left(\frac{a}{p}\right)$ solutions modulo p, since a solution $x \not\equiv 0$ comes with the different solution $-x$.

Theorem 1.1.
Exactly $\frac{p-1}{2}$ of the numbers $1, 2, \ldots, p-1$ are quadratic residues, namely the residues of $1^2, 2^2, \ldots, \left(\frac{p-1}{2}\right)^2$.

Proof. We have $x^2 \equiv y^2$ if and only if $p \mid (x-y)(x+y)$, that is, $x \equiv \pm y$. So x and $p-x$ give the same square and the listed squares are distinct. □

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Theorem 1.2 (Euler's criterion).
If $p \nmid a$ then $a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}$. Consequently $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$ for all integers a, b.

Proof. Let $t = a^{(p-1)/2}$. By Fermat $t^2 \equiv 1$, so $t \equiv \pm 1$. If $a \equiv b^2$ then $t \equiv b^{p-1} \equiv 1$. Hence the $\frac{p-1}{2}$ residues are roots of $x^{(p-1)/2} - 1$, which has at most $\frac{p-1}{2}$ roots modulo p by Lagrange's theorem. So $t \equiv -1$ for a non-residue. For the second statement, both sides are 0 if $p \mid ab$. Otherwise both sides lie in $\{-1, 1\}$ and are congruent to $(ab)^{(p-1)/2}$ modulo p. □

Multiplicativity reduces every symbol to $\left(\frac{-1}{p}\right)$, $\left(\frac{2}{p}\right)$ and $\left(\frac{q}{p}\right)$ for odd primes q. In particular the product of two non-residues is a residue. The proof also gives the polynomial congruence

$$x^{(p-1)/2} - 1 \equiv \prod_r (x - r) \pmod{p},$$

where r runs over the residues in $\{1, \ldots, p-1\}$, because the difference of the two sides has degree less than $\frac{p-1}{2}$ and has $\frac{p-1}{2}$ roots.

Lemma 1.3 (lifting).
Let a be a quadratic residue modulo p. Then $x^2 \equiv a \pmod{p^k}$ has a solution for every $k \geq 1$. Hence, if N is odd and $\gcd(a, N) = 1$, then a is a square modulo N if and only if a is a residue modulo every prime factor of N.

Proof. Suppose $x^2 = a + tp^k$. Then $(x + sp^k)^2 \equiv a + (t + 2xs)p^k \pmod{p^{k+1}}$, and $2x$ is invertible modulo p, so some s makes $t + 2xs$ divisible by p. This is the induction step. The second statement follows from the Chinese remainder theorem. □

1.2 The symbols of -1 and 2

Theorem 2.1.
$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$, so -1 is a residue if and only if $p \equiv 1 \pmod{4}$.

Theorem 2.2. $\left(\frac{2}{p}\right) = 1$ if and only if $p \equiv \pm 1 \pmod{8}$. Hence $\left(\frac{-2}{p}\right) = 1$ if and only if $p \equiv 1$ or $3 \pmod{8}$.

Theorem 2.1 is Euler's criterion with $a = -1$. This reproves the fact from L4-NT1 that odd prime factors of $n^2 + 1$ are 1 modulo 4.

Proof of Theorem 2.2. Let $h = \frac{p-1}{2}$. The product of the even numbers $2, 4, \ldots, 2h$ is $2^h h!$. Replace each even number $2j > h$ by $2j - p$, which is negative with odd absolute value $p - 2j \leq h$. The numbers $2j \leq h$ are the even numbers in $[1, h]$, and the values $p - 2j$ are distinct odd numbers in $[1, h]$. There are $m = h - \lfloor h/2 \rfloor$ of them, which is the number of odd numbers in $[1, h]$. So the absolute values are $1, 2, \ldots, h$ in some order, and

$$2^h h! \equiv (-1)^m h! \pmod{p}.$$

Cancelling $h!$ and using Euler's criterion, $\left(\frac{2}{p}\right) = (-1)^m$. For $p = 8k+1, 8k+3, 8k+5, 8k+7$ we get $m = 2k$, $2k+1$, $2k+1$, $2k+2$. The statement about -2 follows from multiplicativity and Theorem 2.1. □

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Two consequences are worth remembering. For $k \geq 2$, every prime factor p of $2^{2^k}+1$ satisfies $p \equiv 1 \pmod{2^{k+2}}$. Indeed $\operatorname{ord}_p(2) = 2^{k+1}$, so $p \equiv 1 \pmod 8$, so $2^{(p-1)/2} \equiv 1$ and $2^{k+1} \mid \frac{p-1}{2}$. (Euler's factor 641 of $2^{32}+1$ is $5 \cdot 2^7 + 1$.) Secondly, if p and $q = 4p+1$ are both odd primes, then $q \equiv 5 \pmod 8$, so $2^{2p} = 2^{(q-1)/2} \equiv -1 \pmod q$. The order of 2 modulo q divides $4p$ but not $2p$, and is not 4 because $q \nmid 15$. Hence 2 is a primitive root modulo q.

Example 1.
Find all positive integers n for which there is an integer m such that 2^n+1 divides m^2+2.

Solution.
Let $N = 2^n+1$, which is odd. By Lemma 1.3 a suitable m exists if and only if -2 is a residue modulo every prime factor of N. By Theorem 2.2 this means that every prime factor of N is 1 or 3 modulo 8.
If n is odd, the opening paragraph shows that -2 is a square modulo every prime factor of N. So every odd n works.
If $n \equiv 2 \pmod 4$, write $n = 2k$ with k odd. Then $N = 4^k+1 \equiv (-1)^k+1 \equiv 0 \pmod 5$, and 5 is neither 1 nor 3 modulo 8. So these n fail.
If $4 \mid n$, write $n = 2^s t$ with t odd and $s \geq 2$, and let p be a prime factor of N. From $2^n \equiv -1 \pmod p$ the order d of 2 modulo p divides $2^{s+1}t$ but not $2^s t$. Hence $2^{s+1} \mid d$, and $d \mid p-1$ gives $p \equiv 1 \pmod 8$. So these n work.
The answer is all n with $n \not\equiv 2 \pmod 4$.
Remark. A square modulo N is a square modulo each prime factor of N, so we translate the condition into residues before looking at n. The values $N = 3, 5, 9, 17, 33, 65$ for $n \leq 6$ already suggest the answer and the role of the prime 5.

1.3 Quadratic reciprocity and the Jacobi symbol

Theorem 3.1 (quadratic reciprocity).
For distinct odd primes p and q,

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

That is, $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ unless $p \equiv q \equiv 3 \pmod 4$, in which case $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$.

I will not prove this here. The standard proof extends the argument of Theorem 2.2 (Gauss's lemma) and then counts lattice points in a rectangle. See Ireland and Rosen, *A Classical Introduction to Modern Number Theory*, Chapter 5. In an olympiad you may quote every theorem of Sections 1 to 3 without proof.

To evaluate $\left(\frac{a}{p}\right)$, factorise a, flip each odd prime with reciprocity, reduce and repeat. For example $\left(\frac{5}{2029}\right) = \left(\frac{2029}{5}\right) = \left(\frac{4}{5}\right) = 1$. Three formulas, for odd primes p not dividing the top, are worth remembering.

- $\left(\frac{3}{p}\right) = (-1)^{(p-1)/2}\left(\frac{p}{3}\right)$, which is 1 if and only if $p \equiv \pm 1 \pmod{12}$.
- $\left(\frac{-3}{p}\right) = \left(\frac{p}{3}\right)$, which is 1 if and only if $p \equiv 1 \pmod 3$.
- $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$, which is 1 if and only if $p \equiv \pm 1 \pmod 5$.

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Theorem 3.2.
Let $f(n) = an^2 + bn + c$ have integer coefficients and discriminant $D = b^2 - 4ac$, and let $p \nmid a$. Then p divides $f(n)$ for some integer n if and only if $\left(\frac{D}{p}\right) \neq -1$.

Proof. We have $4af(n) = (2an + b)^2 - D$, and $2an + b$ runs through all residues modulo p as n does. □

For $n^2 + n + 2$ the discriminant is -7, and reciprocity gives $\left(\frac{-7}{p}\right) = \left(\frac{p}{7}\right)$. So the odd primes dividing some value of $n^2 + n + 2$ are 7 and the primes congruent to $1, 2, 4$ modulo 7.

Theorem 3.3 (Jacobi symbol).
For an odd positive integer $n = p_1 p_2 \cdots p_k$, a product of primes that need not be distinct, define $\left(\frac{a}{n}\right) = \prod_i \left(\frac{a}{p_i}\right)$, with $\left(\frac{a}{1}\right) = 1$. Let m and n be odd positive integers.
(1) The symbol is multiplicative in a and in n, and depends only on a modulo n.
(2) $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}$ and $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$.
(3) If $\gcd(m, n) = 1$, then $\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}$.

Proof. Statement (1) is in the definition. For odd m, n we have $\frac{mn-1}{2} \equiv \frac{m-1}{2} + \frac{n-1}{2} \pmod 2$, because $4 \mid (m-1)(n-1)$, and $\frac{m^2n^2-1}{8} \equiv \frac{m^2-1}{8} + \frac{n^2-1}{8} \pmod 2$, because $64 \mid (m^2-1)(n^2-1)$. So in (2) and (3) both sides are multiplicative in n, and in (3) also in m. For primes the statements are Theorems 2.1, 2.2 and 3.1. □

If $\gcd(a, n) = 1$ and a is a square modulo n, then $\left(\frac{a}{n}\right) = 1$. The converse is false, as $\left(\frac{2}{15}\right) = \left(\frac{2}{3}\right)\left(\frac{2}{5}\right) = 1$ but 2 is not a square modulo 15. We use the symbol in the other direction. If $\left(\frac{a}{n}\right) = -1$, then a is a non-residue modulo some prime factor of n, although we cannot say which one.

Theorem 3.4.
If the integer a is not a perfect square, then a is a non-residue modulo some odd prime.

Proof. Write $a = dm^2$ with $d \neq 1$ squarefree. We choose a positive integer n by the Chinese remainder theorem. In every case let $n \equiv 1$ modulo each odd prime factor of m that does not divide d. If $d = -1$, let $n \equiv 3 \pmod 4$. If $d = \pm 2$, let $n \equiv 5 \pmod 8$. Otherwise d has an odd prime factor q, and we let $n \equiv 1 \pmod 8$, let n be congruent to a non-residue modulo q, and let $n \equiv 1$ modulo the other odd prime factors of d. In the first two cases $\left(\frac{d}{n}\right) = -1$ by Theorem 3.3(2). In the third, $\left(\frac{-1}{n}\right) = \left(\frac{2}{n}\right) = 1$ and $\left(\frac{\ell}{n}\right) = \left(\frac{n}{\ell}\right)$ for each odd prime $\ell \mid d$, which is -1 for $\ell = q$ and 1 otherwise. So again $\left(\frac{d}{n}\right) = -1$. Hence some prime factor p of n has $\left(\frac{d}{p}\right) = -1$. By the choice of n, $p \nmid m$, so $\left(\frac{a}{p}\right) = -1$. □

Example 2.
Let a, b and n be positive integers. Prove that $4ab - 1$ does not divide $a^n + 1$.
Solution.
Let $N = 4ab - 1$, so $N \geq 3$, $N \equiv 3 \pmod 4$ and $\gcd(a, N) = 1$. Suppose that $N \mid a^n + 1$.
If n is even, note that N has a prime factor $q \equiv 3 \pmod 4$, since a product of primes that are 1 modulo 4 is 1 modulo 4. Then $\left(a^{n/2}\right)^2 \equiv -1 \pmod q$, contradicting Theorem 2.1.
If n is odd, then $-a \equiv a^{n+1} = \left(a^{(n+1)/2}\right)^2 \pmod N$. So $-a$ is a residue modulo every

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prime factor of N, and $\left(\frac{-a}{N}\right) = 1$. We now compute this Jacobi symbol by reciprocity. Write $a = 2^s c$ with c odd. Since $N \equiv 3 \pmod 4$ and $N \equiv -1 \pmod c$, Theorem 3.3 gives

$$\left(\frac{c}{N}\right) = (-1)^{(c-1)/2}\left(\frac{N}{c}\right) = (-1)^{(c-1)/2}\left(\frac{-1}{c}\right) = 1.$$

If $s \geq 1$ then $8 \mid 4ab$, so $N \equiv 7 \pmod 8$ and $\left(\frac{2}{N}\right) = 1$. In both cases $\left(\frac{a}{N}\right) = \left(\frac{2}{N}\right)^s \left(\frac{c}{N}\right) = 1$, and therefore

$$\left(\frac{-a}{N}\right) = \left(\frac{-1}{N}\right)\left(\frac{a}{N}\right) = -1.$$

This is a contradiction.

Remark. For odd n no particular prime factor of N can be named, so Legendre symbols alone give nothing. The Jacobi symbol shows that some prime factor fails. The computation used only $N \equiv -1 \pmod{4a}$, and proves a fact of Euler. If $\gcd(x, y) = 1$, then $x^2 + ay^2$ has no positive divisor congruent to -1 modulo $4a$.

1.4 Thue's lemma and sums of two squares

Lemma 4.1 (Thue).
For every integer r there are integers u, v, not both zero, with $|u|, |v| < \sqrt{p}$ and $u \equiv rv \pmod p$.

Proof. Let $m = \lfloor \sqrt{p} \rfloor$, so $m < \sqrt{p}$. There are $(m+1)^2 > p$ pairs (a, b) with $0 \leq a, b \leq m$, so two different pairs give the same residue of $a - rb$, say $a - rb \equiv a' - rb'$. Put $u = a - a'$ and $v = b - b'$. Then $|u|, |v| \leq m$ and $u \equiv rv$. □

Theorem 4.2 (Fermat).
Every prime $p \equiv 1 \pmod 4$ is a sum of two squares.

Lemma 4.3. If $q \equiv 3 \pmod 4$ is prime and $q \mid x^2 + y^2$, then $q \mid x$ and $q \mid y$.

Theorem 4.4. A positive integer n is a sum of two squares if and only if every prime $q \equiv 3 \pmod 4$ divides n to an even power.

Proof of 4.2. By Theorem 2.1 there is r with $r^2 \equiv -1 \pmod p$. Take u, v as in Thue's lemma. Then $u^2 + v^2 \equiv v^2(r^2 + 1) \equiv 0 \pmod p$ and $0 < u^2 + v^2 < 2p$. Hence $u^2 + v^2 = p$. □

The same argument with $r^2 \equiv -2$ gives $0 < u^2 + 2v^2 < 3p$. If $u^2 + 2v^2 = 2p$, then u is even and $v^2 + 2(u/2)^2 = p$. So every prime $p \equiv 1$ or $3 \pmod 8$ is of the form $x^2 + 2y^2$. For larger coefficients more multiples of p have to be ruled out.

Proof of 4.3. If $q \nmid y$ then $(xy^{-1})^2 \equiv -1 \pmod q$, contradicting Theorem 2.1. So $q \mid y$, and then $q \mid x$. □

Proof of 4.4. If the condition holds, n is a product of factors $2 = 1^2 + 1^2$, primes $p \equiv 1 \pmod 4$ and squares $q^2 = q^2 + 0^2$, and the identity $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$ shows that a product of sums of two squares is a sum of two squares. Conversely suppose $n = x^2 + y^2$ and $q \equiv 3 \pmod 4$ divides n. By Lemma 4.3, $q \mid x, y$, so $n/q^2 = (x/q)^2 + (y/q)^2$ is a sum of two squares in which the exponent of q is smaller by 2. Repeating, we reach exponent 0 or 1, and 1 is impossible by the same lemma. □

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Example 3.
Prove that the equation $y^2 = x^3 + 23$ has no integer solutions.

Solution.
If x is even then $y^2 \equiv 7 \pmod 8$, which is impossible. If $x \equiv 3 \pmod 4$ then $y^2 \equiv 27 + 23 \equiv 2 \pmod 4$, which is impossible. The case $x \equiv 1 \pmod 4$ survives modulo 8, as $x = 1$ gives 24.
Add 4 to both sides, so that

$$y^2 + 2^2 = x^3 + 27 = (x+3)(x^2 - 3x + 9).$$

Suppose $x \equiv 1 \pmod 4$. Then $x^2 - 3x + 9 \equiv 3 \pmod 4$, and $x^2 - 3x + 9 = \left(x - \frac{3}{2}\right)^2 + \frac{27}{4} > 0$. A positive integer that is 3 modulo 4 has a prime factor $q \equiv 3 \pmod 4$. Then $q \mid y^2 + 2^2$, and Lemma 4.3 gives $q \mid 2$, a contradiction.
Remark. After the reductions modulo 4 and 8 we want a prime $q \equiv 3 \pmod 4$ dividing a sum of two squares, one of which is coprime to q. So we add the constant that makes the left side $y^2 + 2^2$ and lets the right side factorise. The positivity of $x^2 - 3x + 9$ must be checked, since -1 is 3 modulo 4 and has no prime factor.

1.5 Counting with the symbol

By Theorem 1.1, $\sum_{x=0}^{p-1} \left(\frac{x}{p}\right) = 0$. The sum of the symbol over a quadratic is as simple.

Lemma 5.1.
Let $f(x) = ax^2 + bx + c$ with $p \nmid a$ and $p \nmid b^2 - 4ac$. Then $\displaystyle\sum_{x=0}^{p-1} \left(\frac{f(x)}{p}\right) = -\left(\frac{a}{p}\right)$.

Proof. Let $D = b^2 - 4ac$. Since $4af(x) = (2ax+b)^2 - D$ and $y = 2ax + b$ runs over all residues, the sum equals $\left(\frac{a}{p}\right)\sum_y \left(\frac{y^2 - D}{p}\right)$. Count the solutions of $y^2 - z^2 \equiv D$ in two ways. Writing $s = y - z$ and $t = y + z$, we need $st \equiv D$, and there are $p - 1$ solutions, one for each $s \not\equiv 0$. On the other hand, for each y there are $1 + \left(\frac{y^2 - D}{p}\right)$ values of z. Hence $p + \sum_y \left(\frac{y^2 - D}{p}\right) = p - 1$. □

Example 4.
How many integers n with $1 \le n \le p - 2$ are such that n and $n + 1$ are both quadratic residues modulo p?

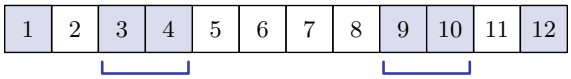
Solution.
For $p \nmid a$ the quantity $\frac{1}{2}\left(1 + \left(\frac{a}{p}\right)\right)$ is 1 for a residue and 0 for a non-residue. So the required number is

$$N = \frac{1}{4}\sum_{n=1}^{p-2} \left(1 + \left(\frac{n}{p}\right)\right)\left(1 + \left(\frac{n+1}{p}\right)\right).$$

Expand into four sums. The first is $p - 2$. The second is $\sum_{n=1}^{p-2} \left(\frac{n}{p}\right) = -\left(\frac{-1}{p}\right)$, and the third is $\sum_{m=2}^{p-1} \left(\frac{m}{p}\right) = -1$. In the fourth, $\sum_{n=1}^{p-2} \left(\frac{n^2+n}{p}\right)$, the missing terms $n = 0$ and $n = p - 1$ are zero, so Lemma 5.1 gives -1.
Therefore $N = \frac{1}{4}\left(p - 4 - \left(\frac{-1}{p}\right)\right)$, that is, $\frac{p-5}{4}$ if $p \equiv 1 \pmod 4$ and $\frac{p-3}{4}$ if $p \equiv 3 \pmod 4$. Modulo 13 the residues are $1, 3, 4, 9, 10, 12$ and the pairs are $(3, 4)$ and $(9, 10)$.

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Remark. The product of the two indicators counts the pairs, and every sum that appears is a sum of the symbol over a polynomial of degree at most 2. A condition on three consecutive numbers leads to a cubic, to which Lemma 5.1 does not apply (Practice 4).



Common traps.

- Euler's criterion and reciprocity are for odd primes. Treat $p = 2$ and the primes dividing the coefficients separately.
- Reciprocity flips only odd numbers. Remove the sign and the powers of 2 first, using Theorems 2.1 and 2.2.
- A Jacobi symbol equal to 1 proves nothing about squares modulo a composite number.
- Lemma 4.3 says that q divides both x and y, not that q cannot divide $x^2 + y^2$. Indeed $3 \mid 3^2 + 6^2$.
- "A number that is 3 modulo 4 has a prime factor that is 3 modulo 4" needs the number to be positive.

1.6 How to spot this in a contest

1. A prime p divides an expression that can be rearranged as "$a \equiv$ square", for instance $p \mid n^2 + k$ or $p \mid 2^n + 1$ with n odd. Translate to $\left(\frac{a}{p}\right) = 1$ and obtain a congruence condition on p.
2. An exponent $\frac{p-1}{2}$ appears. That is Euler's criterion.
3. A divisibility $N \mid x^2 - a$ in which the prime factors of N are unknown. Compute the Jacobi symbol $\left(\frac{a}{N}\right)$. The value -1 is a contradiction.
4. An equation such as $y^2 = x^3 + k$. Look for a constant that gives "sum of two squares = product", then for a factor that is 3 modulo 4.
5. A prime to be written as $x^2 + ky^2$, or small numbers u, v with $p \mid u^2 + kv^2$. Use Thue's lemma.
6. Counting residues with a condition attached. Use indicators and Lemma 5.1.
7. "Infinitely many primes p with $\left(\frac{a}{p}\right) = 1$" needs a Euclid-type argument, see L5-NT2.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆⋆ Let p be a prime with $p \equiv 3 \pmod 4$. The period of the decimal expansion of $\frac{1}{p}$ is the least positive integer k such that p divides $10^k - 1$. Prove that this period is odd if and only if p is congruent to 3, 27, 31 or 39 modulo 40.

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Problem 2 ⋆⋆⋆ Find all primes p such that both $\frac{3^p-1}{2}$ and $\frac{3^{2p+1}-1}{2}$ are prime.

Problem 3 ⋆⋆⋆ Find all primes p for which there is an integer n such that p divides n^4+10n^2+1.

Problem 4 ⋆⋆⋆ Let p be a prime with $p \equiv 3 \pmod 4$. Prove that the number of triples (a,b,c) of integers with $0 \le a,b,c \le p-1$ such that

$$a^2+1 \equiv b^2 \quad \text{and} \quad b^2+1 \equiv c^2 \pmod p$$

is exactly $p-3$.

Problem 5 ⋆⋆⋆⋆ Prove that the equation $y^2 = x^5 - 17$ has no solutions in integers.

Problem 6 ⋆⋆⋆⋆ The sequence $a_0, a_1, a_2, \ldots$ is defined by $a_0 = 2$, $a_1 = 4$ and $a_{n+1} = 4a_n - a_{n-1}$ for $n \ge 1$, so that the sequence begins $2, 4, 14, 52, 194, 724$. Prove that for every odd n, every odd prime factor of a_n is congruent to 1 modulo 12.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT BMO1 2020/21 Problem 6** ⋆⋆
   Pin down the possible Mersenne primes from the first condition, then use the criterion for sums of two squares.
2. **IMO 1996 Problem 4** ⋆⋆
   Combine the two squares to get a sum of two fourth powers divisible by 481 = 13 x 37, then ask which primes can divide such a sum.
3. **EGMO 2016 Problem 6** ⋆⋆⋆
   Reduce to a few Pell-type equations, then compare with the squares modulo 7.
4. **UKMT BMO2 2015/16 Problem 4** ⋆⋆⋆
   The condition says $u^2 + v^2 = 2p^2$. Rewrite the equation as a sum of two squares equal to a square and use what you know about representations of p.
5. **UKMT BMO2 2013/14 Problem 3** ⋆⋆⋆
   For part (b), ask what p dividing a term forces on the previous term, and complete the square.
6. **USAMO 2008 Problem 1** ⋆⋆⋆
   Aim for a product of the form $a^2 + a + 1$. Which primes divide such numbers?
7. **APMO 2014 Problem 3** ⋆⋆⋆
   Reduce to primes. Two different residues with the same value satisfy a quadratic relation. Complete the square and count squares modulo p.
8. **IMO 2008 Problem 3** ⋆⋆⋆⋆
   Work backwards from a large prime p = 1 mod 4 and the square roots of -1 modulo p.
9. **USAMO 2020 Problem 3** ⋆⋆⋆⋆
   Parametrise the a for which a and 4 - a have prescribed residue behaviour, treating p mod 4 separately.

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10. **USAMO 2026 Problem 6** ⋆⋆⋆⋆

Use the parity of the totient to restrict the shape of ab + 1. Reciprocity for -3 appears in one case.

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4 Hints

1. If $10^k \equiv 1$ with k odd, then $10 \equiv 10^{k+1}$ is a square. For the converse look at the exponent $\frac{p-1}{2}$.
2. First show that $q = 2p+1$ must be prime. Then $3^{2p} \equiv 1 \pmod q$, and the question is which of $3^p - 1$ and $3^p + 1$ is divisible by q.
3. Write $n^4 + 10n^2 + 1$ as a square plus a multiple of n^2 in two different ways. For the converse, try $n = a + b$ with a^2 and b^2 suitably chosen.
4. For each residue n, count the triples with $b^2 \equiv n$. The count is a product of three factors $1 + \left(\frac{\cdot}{p}\right)$. For the sum of $\left(\frac{n^3-n}{p}\right)$, replace n by $-n$.
5. Add 49 to both sides. A factor of the right side is 3 modulo 4 in the case that survives, and Lemma 4.3 leaves only one prime to worry about. Count the powers of that prime.
6. Find a formula for a_{2m+1} in terms of a_m and a_{m+1}, and a quadratic expression in a_m and a_{m+1} that does not depend on m. Then look at $(a_{m+1} \pm a_m)^2$ modulo the prime.

5 Solutions

Problem 1

Let p be a prime with $p \equiv 3 \pmod 4$. The period of the decimal expansion of $\frac{1}{p}$ is the least positive integer k such that p divides $10^k - 1$. Prove that this period is odd if and only if p is congruent to 3, 27, 31 or 39 modulo 40.

Let k be the order of 10 modulo p. Note that $p \neq 2, 5$.

Step 1. The period is odd if and only if 10 is a residue modulo p. If k is odd, then $10 \equiv 10^{k+1} = \left(10^{(k+1)/2}\right)^2 \pmod p$, so 10 is a residue. Conversely, if 10 is a residue, then $10^{(p-1)/2} \equiv 1 \pmod p$ by Euler's criterion, so k divides $\frac{p-1}{2}$, which is odd because $p \equiv 3 \pmod 4$. Hence k is odd.

Step 2. The symbol. We have $\left(\frac{10}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{5}{p}\right)$, so 10 is a residue if and only if the two symbols on the right are equal. Since $p \equiv 3 \pmod 4$, Theorem 2.2 gives $\left(\frac{2}{p}\right) = 1$ for $p \equiv 7 \pmod 8$ and $\left(\frac{2}{p}\right) = -1$ for $p \equiv 3 \pmod 8$. By reciprocity $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$, which is 1 for $p \equiv \pm 1 \pmod 5$ and -1 for $p \equiv \pm 2 \pmod 5$.

Both symbols are 1 when $p \equiv 7 \pmod 8$ and $p \equiv \pm 1 \pmod 5$. Among $7, 15, 23, 31, 39$ these are 31 and 39. Both symbols are -1 when $p \equiv 3 \pmod 8$ and $p \equiv \pm 2 \pmod 5$. Among $3, 11, 19, 27, 35$ these are 3 and 27. So the period is odd if and only if $p \equiv 3, 27, 31, 39 \pmod{40}$. □

Remark. An odd exponent k with $10^k \equiv 1$ can be raised by one to make an even exponent, and this is the same step as in the opening of the lesson. For example $\frac{1}{31}$ has period 15 and $\frac{1}{43}$ has period 21, while $\frac{1}{7}$ has period 6. For $p \equiv 1 \pmod 4$ the converse in Step 1 fails, as $\frac{1}{13}$ has period 6 although $10 \equiv 6^2 \pmod{13}$.

Problem 2

Find all primes p such that both $\frac{3^p-1}{2}$ and $\frac{3^{2p+1}-1}{2}$ are prime.

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Write $y(n) = \frac{3^n-1}{2}$ and $q = 2p+1$. For $p = 2$ we get $y(2) = 4$, which is not prime, so let p be odd.

Step 1. q is prime. If $q = uv$ with $1 < u < q$, then $u \geq 3$ and $3^u - 1$ divides $3^q - 1$, so $y(u)$ divides $y(q)$ with $1 < y(u) < y(q)$, and $y(q)$ is not prime.

Step 2. The case $p \neq 3$. If $p \equiv 1 \pmod 3$ then $3 \mid q$ and $q > 3$, contradicting Step 1. So $p \equiv 2 \pmod 3$ and $q \equiv 2 \pmod 3$. Also $q = 2p+1 \equiv 3 \pmod 4$ because p is odd. By Euler's criterion and reciprocity,

$$3^p = 3^{(q-1)/2} \equiv \left(\frac{3}{q}\right) = -\left(\frac{q}{3}\right) = -\left(\frac{2}{3}\right) = 1 \pmod q.$$

So q divides $3^p - 1 = 2y(p)$, and as q is odd, q divides $y(p)$. Since $y(p)$ is prime, $y(p) = q$, that is, $3^p = 4p+3$. This is false, because $3^p > 4p+3$ for all $p \geq 3$ (we have $27 > 15$, and increasing p by 1 triples the left side and adds 4 to the right). So there is no such prime.

Step 3. The case $p = 3$. Here $y(3) = 13$ and $y(7) = 1093$, and both are prime. For 1093 we check that none of the primes up to 31 is a factor, and $37^2 > 1093$.

Hence $p = 3$ is the only such prime. □

Remark. Fermat's theorem gives $q \mid (3^p - 1)(3^p + 1)$ at once, but only the first factor is useful, since q dividing $3^p + 1$ says nothing about $y(p)$. Euler's criterion decides which factor q divides. For $p = 3$ we have $q = 7 \equiv 1 \pmod 3$, so 7 divides $3^3 + 1$, and this is how $p = 3$ survives.

Answer: $p = 3$

Problem 3

Find all primes p for which there is an integer n such that p divides $n^4 + 10n^2 + 1$.

Let $f(n) = n^4 + 10n^2 + 1$. Since $f(1) = 12$, the primes 2 and 3 have the property. From now on let $p \geq 5$.

Necessity. Suppose $p \mid f(n)$. Then $p \nmid n$, because $f(n) \equiv 1 \pmod n$. We have the identities

$$f(n) = (n^2+1)^2 + 2(2n)^2 = (n^2-1)^2 + 3(2n)^2.$$

Hence $(n^2+1)^2 \equiv -2(2n)^2$ and $(n^2-1)^2 \equiv -3(2n)^2 \pmod p$. As $2n$ is invertible modulo p, both -2 and -3 are quadratic residues modulo p. By Theorem 2.2 and the formula for $\left(\frac{-3}{p}\right)$ in Section 3, $p \equiv 1$ or $3 \pmod 8$ and $p \equiv 1 \pmod 3$. So $p \equiv 1$ or $19 \pmod{24}$.

Sufficiency. Let $p \equiv 1$ or $19 \pmod{24}$. Then -2 and -3 are residues, so there are integers a, b with $a^2 \equiv -2$ and $b^2 \equiv -3 \pmod p$. Put $n = a + b$. Then $n^2 \equiv -5 + 2ab$, so $(n^2+5)^2 \equiv 4a^2b^2 \equiv 24$, and

$$f(n) = (n^2+5)^2 - 24 \equiv 0 \pmod p.$$

The required primes are 2, 3 and the primes congruent to 1 or 19 modulo 24. □

Remark. Completing the square in the obvious way gives $(n^2+5)^2 \equiv 24$, which only says that 6 is a residue. The constant term 1 allows two more completions, and each one gives a symbol. The converse comes from reading the first completion backwards, which suggests $n^2 = -5 + 2ab$ with $a^2b^2 = 6$. The smallest cases are $f(3) = 172 = 4 \cdot 43$ and $f(2) = 57 = 3 \cdot 19$.

Answer: $p = 2$, $p = 3$ and all primes $p \equiv 1$ or $19 \pmod{24}$

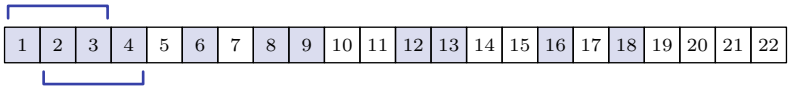
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Problem 4

Let p be a prime with $p \equiv 3 \pmod 4$. Prove that the number of triples (a,b,c) of integers with $0 \le a,b,c \le p-1$ such that

$$a^2+1 \equiv b^2 \quad \textit{and} \quad b^2+1 \equiv c^2 \pmod p$$

is exactly $p-3$.



Write $\chi(t) = \left(\frac{t}{p}\right)$. For a residue n, the triples with $b^2 \equiv n$ are given by a solution a of $a^2 \equiv n-1$, a solution b of $b^2 \equiv n$ and a solution c of $c^2 \equiv n+1$. By Section 1 their number is $\big(1+\chi(n-1)\big)\big(1+\chi(n)\big)\big(1+\chi(n+1)\big)$. So the required number is

$$N = \sum_{n=0}^{p-1} \big(1+\chi(n-1)\big)\big(1+\chi(n)\big)\big(1+\chi(n+1)\big).$$

We expand the product. The constant term contributes p. Each of the three sums of a single symbol is $\sum_t \chi(t) = 0$.

The products of two symbols give the sums of $\chi(n^2-n)$, $\chi(n^2+n)$ and $\chi(n^2-1)$ over all residues n. The discriminants are 1, 1 and 4, so each sum is -1 by Lemma 5.1.

The product of three symbols gives $S = \sum_{n=0}^{p-1} \chi(n^3-n)$. Since $p \equiv 3 \pmod 4$ we have $\chi(-1) = -1$, so $\chi\big((-n)^3-(-n)\big) = -\chi(n^3-n)$. As $n \mapsto -n$ permutes the residues, $S = -S$ and $S = 0$.

Therefore $N = p+0-3+0 = p-3$. □

Remark. Counting by the value of b^2 turns the system into a statement about three consecutive numbers $n-1, n, n+1$. Lemma 5.1 does not reach the cubic n^3-n, but this cubic is an odd function and χ is odd when $p \equiv 3 \pmod 4$, which is why the hypothesis is there. The figure shows $p = 23$, where $n = 2$ and $n = 3$ give 8 triples each and $n = 1$ gives 4, in total 20. For $p = 13$ the cubic sum is -6 and the number of triples is 4.

Problem 5

Prove that the equation $y^2 = x^5 - 17$ has no solutions in integers.

Suppose that (x,y) is a solution.

Step 1. $x \equiv 1 \pmod 4$. If x is even, then $y^2 \equiv -17 \equiv 7 \pmod 8$, which is impossible. If $x \equiv 3 \pmod 4$, then $x^5 \equiv 3$ and $y^2 \equiv 3-17 \equiv 2 \pmod 4$, which is impossible.

Step 2. A factor that is 3 *modulo* 4. Adding 49 gives

$$y^2+7^2 = x^5+2^5 = (x+2)\,Q, \qquad Q = x^4-2x^3+4x^2-8x+16.$$

Since $x \equiv 1 \pmod 4$, we get $Q \equiv 1-2 \equiv 3 \pmod 4$. Also $Q = (x^2-x)^2+3x^2-8x+16 > 0$, because $3x^2-8x+16$ has negative discriminant. A positive integer in which every prime $q \equiv 3$

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$(\mathrm{mod}\ 4)$ appears to an even power is 1 modulo 4. Hence some prime $q \equiv 3 \pmod 4$ divides Q to an odd power.

Step 3. The prime is 7. We have $q \mid y^2 + 7^2$, so $q \mid y$ and $q \mid 7$ by Lemma 4.3. Hence $q = 7$, the exponent of 7 in Q is odd, and $7 \mid y$.

Step 4. Counting powers of 7. Write $y = 7t$. Then $y^2 + 49 = 49(t^2 + 1)$ and $7 \nmid t^2 + 1$ by Theorem 2.1, so the exponent of 7 in $(x + 2)Q$ is exactly 2. As the exponent in Q is odd, this exponent is 1, and 7 divides $x + 2$. But then $x \equiv -2 \pmod 7$ and

$$Q \equiv 16 + 16 + 16 + 16 + 16 = 80 \equiv 3 \pmod 7,$$

so $7 \nmid Q$. This is a contradiction. □

Remark. The constant must turn $x^5 - 17$ into a sum of two fifth powers, and $-17 + 49 = 32$ does this. Unlike Example 3, the added square 7^2 has a prime factor that is 3 modulo 4, so Lemma 4.3 does not finish the proof and the prime 7 has to be followed through both factors. The equation has solutions modulo 4, modulo 7 and modulo 11, so some idea of this kind is needed.

Problem 6

The sequence $a_0, a_1, a_2, \ldots$ is defined by $a_0 = 2$, $a_1 = 4$ and $a_{n+1} = 4a_n - a_{n-1}$ for $n \geq 1$, so that the sequence begins $2, 4, 14, 52, 194, 724$. Prove that for every odd n, every odd prime factor of a_n is congruent to 1 modulo 12.

Let $\alpha = 2 + \sqrt{3}$ and $\beta = 2 - \sqrt{3}$, the roots of $t^2 - 4t + 1$. The sequence $\alpha^n + \beta^n$ satisfies the same recurrence and has the same first two terms 2 and 4, so $a_n = \alpha^n + \beta^n$. Note that $\alpha\beta = 1$ and $\alpha + \beta = 4$.

Step 1. Two identities. For all $m \geq 0$,

$$a_m a_{m+1} = \alpha^{2m+1} + \beta^{2m+1} + (\alpha\beta)^m(\alpha + \beta) = a_{2m+1} + 4.$$

Next let $I_m = a_{m+1}^2 - 4a_{m+1}a_m + a_m^2$. Using $a_{m+2} - 4a_{m+1} = -a_m$,

$$I_{m+1} = a_{m+2}(a_{m+2} - 4a_{m+1}) + a_{m+1}^2 = -(4a_{m+1} - a_m)a_m + a_{m+1}^2 = I_m.$$

Since $I_0 = 16 - 32 + 4 = -12$, we get $a_{m+1}^2 + a_m^2 = 4a_m a_{m+1} - 12$ for all m.

Step 2. Two squares modulo p. Let $n = 2m + 1$ and let p be an odd prime factor of a_n. Put $x = a_m$ and $y = a_{m+1}$. By Step 1, $xy \equiv 4$ and $x^2 + y^2 \equiv 16 - 12 = 4 \pmod p$. Hence

$$(x + y)^2 \equiv 4 + 8 = 12, \qquad (x - y)^2 \equiv 4 - 8 = -4 \pmod p.$$

Step 3. The residue of p. As p is odd, the second congruence gives $\left(\frac{x-y}{2}\right)^2 \equiv -1$, so $p \equiv 1 \pmod 4$ by Theorem 2.1. In particular $p \neq 3$, so the first congruence shows that 3 is a nonzero square modulo p, and $p \equiv \pm 1 \pmod{12}$ by Section 3. Together these give $p \equiv 1 \pmod{12}$. □

Remark. A prime dividing a_n gives one relation, and a residue condition needs a square. The product formula brings in the neighbours a_m, a_{m+1}, and the invariant I_m supplies $x^2 + y^2$, after which $(x \pm y)^2$ are known. For even n the same method gives less. From $a_{2m} = a_m^2 - 2$ we only get that 2 is a residue, and indeed $a_6 = 2 \cdot 7 \cdot 193$. The first odd cases are $a_3 = 4 \cdot 13$, $a_5 = 4 \cdot 181$ and $a_7 = 4 \cdot 2521$.